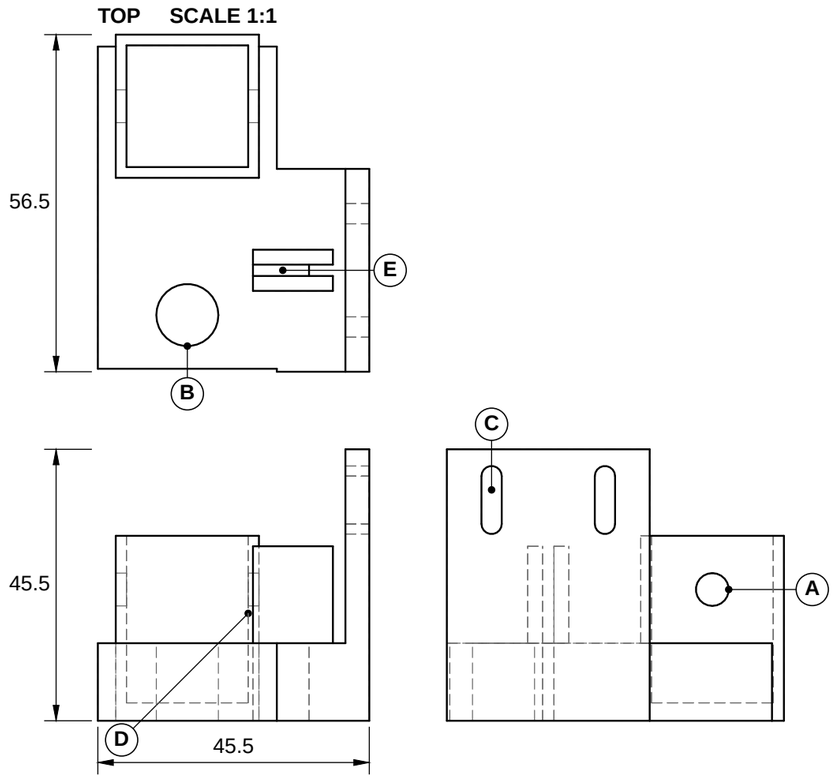
head_foot

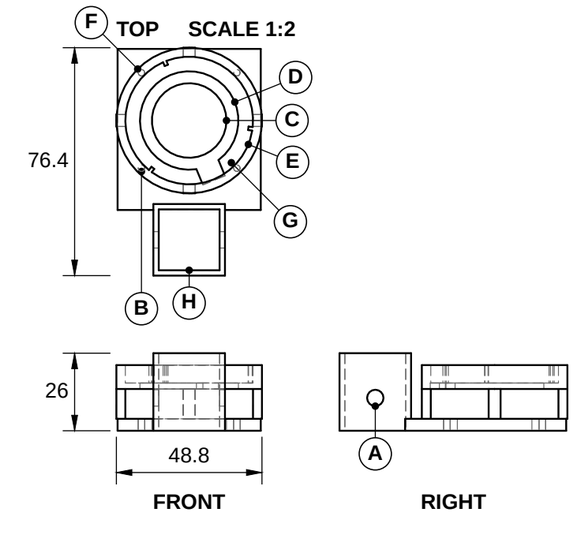
Printed part (P), qty 1, cad/out/printed/head_foot.step. Print: PETG, foot down (as exported), 5 walls, 40% infill; clear the 1.9 mm flag slot and the board slots with a hacksaw blade



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU (2 walls on one axis), axis X: M5 clearance
B	Ø10.4 THRU, axis Z: M10x1 buffer body
C	2x slot 3.4 wide x 11.4 long, axis X, centres 19 apart in Y: endstop board M3 slots (trip adjustment)
D	Drop-arm socket 20.4 square, floor 3 above the underside; cross hole 19 above the floor (match-drill the arm)
E	Flag guide slot 1.9 wide through the fins and the foot (clear it after printing); underside at c + 62

camera_bracket

Printed part (P), qty 1, cad/out/printed/camera_bracket.step. Print: PETG matte black, shelf face down (as exported), 4 walls, 40% infill, brim; the ring seat bridges ~30 mm between its 4 legs (tree supports there if your printer sags)



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU (2 walls on one axis), axis X: M5 clearance
B	4x Ø2.2 THRU, axis Z, on 32 x 32 (X x Y): ELP board, M2 + nuts, about the lens axis ([Guessing], 3.3)
C	Ø25 THRU, axis Z: lens window
D	Ø33 THRU, interrupted, axis Z: ring lip (ring back face rests on it)
E	Ø42.8 x 6 deep, interrupted, axis Z: head LED ring seat bore (ribbed)
F	Ø48.8 x 8 long (outside), axis Z: ring seat tube
G	Ring seat: 3 ribs leave Ø40 (shave for a larger ring, BUILD_GUIDE 3.3); lip 16 below the board shelf
H	Lens axis 40 from the drop-arm centre (ARM_TO_CUP + lens offset HEAD_CAM_DY = -5)

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-12 of D-17**
Printed parts: head	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ 8eab5dd (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**